ROBUST UNBIASED WATERMARKING: FROM AUTOREGRESSIVE TO DIFFUSION MODELS

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Abstract

With the exponential growth in the generative capabilities of Large Language Models (LLMs), the authenticity of digital content faces an unprecedented crisis of trust. To address challenges such as fake news, academic fraud, and copyright infringement, establishing reliable content provenance mechanisms has emerged as a central topic in the field of AI safety. While zero-bit watermarking techniques—which solely detect whether content is machine-generated—have become increasingly mature, multi-bit watermarking techniques capable of embedding specific identity information (such as User IDs) continue to grapple with the "trilemma" (impossible triangle) of generation quality, payload capacity, and robustness. This report details the core outcomes and strategic pivots of this work as a UNSW COMP9991 course project and its subsequent research initiatives.

This phase of the research primarily focused on reproducing and enhancing **BiMark**, a watermarking framework designed for autoregressive models based on Unbiased Multilayer Reweighting. We posited two core hypotheses: that integrating Error Correction Codes (ECC) and entropy-adaptive signal strength modulation (Variable δ) would significantly bolster BiMark's robustness against local editing attacks. However, rigorous experimental validation yielded a counter-intuitive negative result. Within the current autoregressive multi-bit architecture, the naive superposition of ECC diluted the statistical signal strength of the underlying voting mechanism, resulting in reduced extraction accuracy. Furthermore, the dynamic δ strategy proved ineffective as it disrupted synchronization on the detection side.

Driven by the profound insight that the "Pareto frontier" of autoregressive watermarking is approaching saturation, alongside the rise of non-autoregressive architectures in the generative landscape, this report establishes a strategic pivot for the COMP9993 phase: advancing into **watermarking research for Diffusion Language Models (DLMs)**. We aim to investigate how to integrate BiMark's multilayer framework with the "Expectation Boost" mechanism proposed by Gloaguen et al. within the non-sequential process of discrete diffusion generation. Our objective is to construct the first robust multi-bit watermarking protocol tailored for diffusion models.

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Chapter 1

Introduction

The evolutionary trajectory of Artificial Intelligence has decisively shifted from discriminative tasks to generative capabilities, a paradigm shift that is reshaping the global digital information ecosystem. Generative models, particularly Large Language Models (LLMs) exemplified by GPT series [21], Llama [27], and DeepSeek [10], have achieved a level of proficiency where the generation of coherent, context-rich text is virtually indistinguishable from human output. While this technological leap has significantly expedited the automation of code generation, creative writing, and customer service, its "double-edged" nature has precipitated profound ethical and legal concerns regarding the propagation of misinformation, academic dishonesty, phishing campaigns, and intellectual property attribution.

1.1 The Necessity and Evolution of Watermarking Technologies

To mitigate the aforementioned risks, digital watermarking has transitioned from a niche cryptographic application to a central pillar of AI safety. Unlike **Post-hoc Detection** methods—which rely on classifiers to identify statistical anomalies in machine-generated text and often suffer from high false-positive rates and rapid obsolescence with model iteration—watermarking serves as a proactive intervention mechanism. During the **inference-time** generation phase, it directly embeds statistical signals—imperceptible to humans but algorithmically detectable—into the probability distribution of the text.

Regulatory frameworks, such as the European Union's AI Act [5] and Executive Orders in the United States [12], have underscored the necessity of establishing reliable content identification mechanisms. However, simplistic binary labeling (distinguishing "AI-generated" from "human-written" content) proves insufficient in scenarios requiring accountability attribution. To trace the origins of misuse—such as identifying the specific user account responsible for generating hate speech—model deployers urgently require **Multi-bit Watermarking** technologies. These techniques enable the embedding of high-dimensional payloads (e.g., a 32-bit User ID), thereby facilitating a qualitative shift from mere "detection" to precise "attribution."

1.2 The Dominance and Boundaries of the Autoregressive Paradigm

To date, the vast majority of watermarking research has concentrated on **Autoregressive Language Models (ARLMs)**. ARLMs generate tokens strictly in a left-to-right sequence, where the generation probability of the current token is conditioned entirely on the preceding context. Classical watermarking schemes in this

domain, such as KGW [14], exploit this sequential dependency by utilizing historical tokens as seeds for a Pseudo-Random Number Generator (PRNG) to bias the sampling of the subsequent token.

However, the underlying architecture of Generative AI is undergoing a paradigm shift. **Diffusion Language Models (DLMs)**—a non-autoregressive architecture based on iterative denoising processes—are exhibiting superior controllability, parallel generation capabilities, and potential for error correction [31, 8, 18]. DLMs break the rigid constraint where "preceding text determines subsequent text." Their stochastic generation dynamics render traditional ARLM watermarking techniques, which rely on fixed prefix histories, theoretically inapplicable. This technical discontinuity constitutes the most significant gap in current watermarking research.

1.3 Project Scope and Core Objectives

This report documents the research trajectory of the COMP9991 phase, which was dedicated to mastering and advancing beyond the current pinnacle of ARLM watermarking technology: the **BiMark** framework [6]. The research objectives were structured across three dimensions:

1. **Reproduction and Deconstruction:** To rigorously implement BiMark and dissect the mathematical underpinnings of its "Unbiased Reweighting" and "Multilayer Voting" mechanisms.
2. **Innovation and Validation:** To attempt to resolve the robustness bottlenecks of multi-bit watermarking against editing attacks by introducing Reed-Solomon Error Correction Codes (RS-ECC) [23] and dynamic layer signal strength strategy.
3. **Evaluation and Pivot:** Based on definitive experimental conclusions—specifically that within the current architecture, merely increasing coding redundancy paradoxically reduced the Signal-to-Noise Ratio (SNR)—to substantiate the necessity of pivoting towards research on watermarking for diffusion models.

Chapter 2

Literature Review

This chapter provides a comprehensive and in-depth survey of the text watermarking landscape. Beginning with early linguistically inspired heuristics, we critically analyze the evolutionary logic of **Logits-based intervention watermarking**, elucidating the technical leaps achieved in **unbiased** and **multi-bit** watermarking schemes. The review culminates by addressing the novel challenges introduced by the emergence of **Diffusion Models**. The objective of this survey is to demonstrate a holistic mastery of the field and to provide a solid theoretical underpinning for the methodological choices detailed in subsequent chapters.

2.1 Early Text Watermarking: Explorations in the Pre-Neural Era

Prior to the dominance of the Transformer architecture in NLP, text watermarking primarily relied on explicit modifications to text formatting or linguistic structures. While these methodologies laid the groundwork for **steganography**, they prove exceptionally fragile within modern adversarial environments.

2.1.1 Format-Based Watermarking

Format-based techniques aim to embed information by altering the visual presentation of the text rather than its semantic content. The fatal flaw of this category lies in its inherent **Fragility**. These methods are defenseless against **"Canonicalization Attacks"**; an attacker need only pass the text through an ASCII filter or reformat it to obliterate the watermark. Consequently, they are rendered unsuitable as API-level regulatory tools.

- **Line/Word-Shift Coding:** Early methods proposed by Brassil et al. encode binary information by fine-tuning the vertical or horizontal spacing of lines or words. While effective for physical printed documents, in digital environments, simple "copy-paste" operations or format conversions (e.g., from PDF to plain text) can completely eradicate the watermark signal [1].
- **Unicode Injection:** Modern variants, such as "UniSpaCh" or "EasyMark," exploit **Zero-Width Spaces** [19] or **Homoglyphs** (e.g., the Cyrillic 'a' versus the Latin 'a') to conceal information [24].

2.1.2 Lexical and Syntactic Substitution

To counter formatting attacks, researchers turned to direct modification of the textual content. However, these methods confront severe **quality bottlenecks**. Rule-based substitutions lack the context-awareness of neural models, often resulting in semantic rigidity or ambiguity. More critically, they are highly susceptible to **Paraphrasing Attacks**; when an attacker employs an LLM to polish the text, the specific lexical choices carrying the watermark are often completely reconstructed.

- **Synonym Substitution:** Relying on lexical databases like WordNet, specific words are replaced with synonyms based on a secret key (e.g., swapping "large" for "big" to encode a '1') [26].
- **Syntactic Restructuring:** Utilizing syntactic parse trees to embed information by transforming sentence structures (e.g., converting active voice to passive voice or shifting adverbial prepositions) [17].

2.2 The Rise of Logits-based Watermarking: From KGW to Unbiased Schemes

The emergence of LLMs has rendered **Inference-time Intervention** a viable strategy. Rather than resorting to post-generation modification, this approach involves directly manipulating the logits—the unnormalized probabilities of the next token—during the generation process. This allows the watermark to be embedded directly within the statistical distribution of the output.

2.2.1 Foundational Work: KGW (Soft Red List)

The KGW [14] scheme proposed by Kirchenbauer et al., also known as the **Soft Red List**, serves as a seminal milestone in this field. By leveraging the inherent randomness in the sampling process, this scheme provides a statistical method to detect generated text without requiring access to model parameters.

The core mechanism of the scheme operates at each generation step t. Based on the hash value of the preceding context $h(x_{t-k}, \ldots, x_{t-1})$, the vocabulary V is pseudo-randomly partitioned into a **"Green List" (G)** and a **"Red List" (R)**. By applying a positive bias (δ) to the logits of tokens in the Green List, the model is induced to sample Green List tokens at a frequency significantly higher than random chance.

The detection phase verifies the presence of the watermark by analyzing the density of Green List tokens via a Z-test. For human-written text, the probability of a token falling into the Green List is approximately 50% (random chance). However, for watermarked text, the presence of δ significantly elevates this ratio. By calculating the **Z-score**, one can establish with extremely high confidence (and an extremely low p-value) that the text originates from a machine.

Although the KGW scheme successfully validated the feasibility of watermarking, it exposed a fundamental trade-off: increasing the watermark strength (i.e., increasing δ) improves detection rates but simultaneously forces the model to select suboptimal tokens. This results in a significant degradation of text quality, often measured by increased Perplexity (PPL). Furthermore, the scheme demonstrates limited embedding capability for low-entropy text (such as code or factual statements) and remains vulnerable to **Paraphrasing Attacks**.

2.2.2 Enhancing Robustness: Decoupling Context Dependencies

Attackers have discovered that modifying preceding tokens alters the hash seed, thereby causing the detector to become **desynchronized**. Addressing these limitations in robustness, subsequent research introduced the **Unigram-Watermark** [33] scheme. This work demonstrated that hash mechanisms relying on long contexts are highly vulnerable to text editing. Consequently, the Unigram approach

simplified context dependency to a single token ($K = 1$) or employed a global hashing strategy.

The theoretical contribution of this approach lies in proving that fixed Green/Red partitioning or minimizing context dependency significantly enhances robustness against **Local Edits** and synonym substitutions. However, this gain in robustness comes at the expense of security—fixed partition patterns are more susceptible to being **reverse-engineered** by attackers via statistical analysis. Additionally, researchers have proposed schemes such as **PatternMark** [3], which attempt to construct sequence-independent watermark patterns to resist reordering attacks, though this often requires a sacrifice in embedding capacity.

2.2.3 Pursuing Distribution Fidelity: Unbiased Watermarking and the Quality Trade-off

Traditional Logit-bias methods (such as KGW [14] and Unigram [33]) inevitably distort the original probability distribution of Language Models by forcibly elevating the probabilities of specific tokens. This distortion directly precipitates a rise in text Perplexity (PPL), signifying a degradation in generation quality. To address the quality deterioration introduced by watermarking, the academic community introduced the concept of **"Unbiased"** or **"Distortion-free"** watermarking. The core objective of these methodologies is to ensure that the probability distribution of the watermarked text aligns perfectly with that of the original model **in expectation**, thereby theoretically eliminating any negative impact on generation quality.

The **α-reweight** and **γ-reweight** schemes [13] (**Unbiased Reweighting**) proposed by Hu et al. exemplify this direction. Unlike KGW, which directly applies a logit bias, this method introduces a reweighting function R that satisfies the condition $\mathbb{E} = P$. This method defines unbiased watermarking from a theoretical standpoint: it mandates that the distribution of the watermarked model, after **Marginalization** over the secret key, must be strictly equivalent to the original model's distribution. This implies that for an observer without access to the key, the statistical characteristics of the watermarked text are indistinguishable from the original, thus achieving **"Stealthiness."** This property is critical for application scenarios that demand not only content authenticity but also the preservation of the original generation style without perceptible alteration.

Building upon this foundation, Wu et al. proposed the **DiPmark** framework [29], which explicitly establishes **"Distribution-preserving," "Accessible,"** and **"Resilient"** as its three primary design objectives. DiPmark ensures distributional consistency during generation via a reweighting strategy based on vocabulary permutation. A significant contribution of this work is the proposal of a novel detection statistic—the **"Green Token Ratio."** By designing a reweighting strategy in the form of a **convex combination** (DiP-reweight), the method ensures that the elevation of the sum of Green List token probabilities does not disrupt the expectation of the overall distribution. More crucially, Wu et al. provided a rigorous **"Certified Radius"** analysis for DiPmark's detection algorithm, quantifying the maximum edit distance the watermark can withstand while keeping the False Positive Rate (FPR) controlled. Experimental results demonstrate that in high-entropy tasks

such as machine translation, DiPmark significantly reduces text quality degradation compared to traditional Soft Watermarks.

In the evolutionary path of unbiased watermarking, the **"MCMARK"** scheme [4] proposed by Chen et al. represents an innovation in **Multi-Channel** strategies. MCMARK abandons binary Red-Green partitioning in favor of dividing the vocabulary into l **Segments**. During the generation process, the watermark key randomly selects one segment for probability boosting. The detection process is then transformed into verifying whether the generated tokens fall into the segment indicated by the key. Theoretical analysis by Chen et al. indicates that in the special case where $l = 2$, MCMARK achieves a lower **"True-Negative Rate"**—meaning the watermark signal is harder to miss—compared to baseline methods like DiPmark [29] and STA-1 [16]. By modeling the watermarking problem as a **constrained optimization problem**, MCMARK maximizes the probability mass of the selected segment while maintaining unbiasedness, thereby significantly enhancing detection efficacy. This study reveals a trade-off relationship between the channel count l and robustness: while increasing the number of channels improves the granularity of the detection signal, it simultaneously reduces robustness against tampering.

2.3 The Evolution of Multi-bit Watermarking: From Detection to Attribution

While zero-bit watermarking addresses the fundamental question of "whether content is AI-generated," multi-bit watermarking is designed to resolve the specific question of "who generated it." The core technical challenge lies in embedding a message m comprising b bits within a text sequence of limited length.

2.3.1 Naive Enumeration and the Blocking Dilemma

Early attempts at multi-bit watermarking were constrained by either computational complexity or degradation in text quality. Wang et al. and Fernandez et al. introduced **CTWL** [28], a scheme that incorporates the message m as a component of the hash seed. During detection, the system must iterate through all possible values of m to identify the message that yields the highest Green List matching rate. The computational complexity of this approach is $O(2^b)$, rendering it computationally infeasible for long messages (e.g., a 32-bit ID).

Separately, Yoo et al. proposed the **MPAC** framework [32]. This method partitions the vocabulary into 2^d subsets, with each subset corresponding to a specific message segment. However, to embed information effectively, the Green List ratio γ must be fixed at $1/2^d$. As d increases, γ diminishes rapidly, resulting in a precipitous decline in generation quality due to the scarcity of candidate tokens.

2.3.2 Breakthroughs in Modern Multi-bit Architectures

To break the impasse between capacity and quality, a variety of innovative architectures have recently emerged.

The **RSBH** scheme [20] proposed by Qu et al. achieves robust multi-bit embedding by unifying the concepts of message enumeration and bit allocation. They identified a severe **"Frequency Imbalance"** issue in existing **Bit Assignment** methods: due to the highly uneven frequency of tokens in natural language (e.g.,

"the" appears much more frequently than other words), certain bit positions are encoded frequently while others lack sufficient "carriers," leading to decoding errors. To address this, unlike fragile bit-wise embedding, RSBH partitions the message into segments and pseudo-randomly assigns these segments to tokens. The core contribution of this work lies in the integration of **Reed-Solomon (RS) Codes**, theoretically proving that the message can be perfectly recovered provided the edit distance remains within a certain bound. This established the necessity of Error Correction Codes (ECC) in high-capacity watermarking. However, as RSBH remains founded on the biased KGW framework, it inevitably suffers from quality degradation.

Similarly, the **RBC** scheme [2] proposed by Chao et al. reframes the watermarking problem as a **binary channel transmission** task. By mapping tokens to binary strings and applying a **Correlated Binary Sampling Channel (CBSC)**, RBC utilizes **Low-Density Parity-Check (LDPC)** codes to achieve distortion-free embedding. Compared to simple redundant coding, LDPC codes offer superior error correction capabilities, enabling RBC to withstand significant deletion and editing attacks. This represents an important theoretical breakthrough, demonstrating the potential of combining ECC with unbiasedness. Chao et al. demonstrated that, compared to simple mapping or distributional shifting, introducing advanced ECC (such as LDPC) allows for high-strength detection within extremely short text lengths, significantly boosting the watermark's **"Bit Rate."** However, despite employing channel coding techniques, the original RBC framework was explicitly designed as a **Zero-bit detection scheme**. Its embedded message m is pseudo-randomly generated based on the preceding context for the purpose of self-verification, rather than being derived from extrinsic user-supplied information (such as a User ID). Consequently, the detection algorithm described in the paper yields a binary classification result (Watermark Present/Absent), rather than extracting a specific bit sequence.

The **MajorMark** scheme [30] proposed by Xu et al. approaches the problem from the perspective of coding strategy optimization. They observed an inherent contradiction between quality and robustness when using a fixed Green List ratio γ (e.g., set to 0.5). MajorMark utilizes the **"Majority Bit"** within the message to guide the construction of the Green List, dynamically ensuring that the Green List ratio γ remains greater than or equal to 0.5. This design ingeniously leverages a larger sampling space to enhance text quality. Furthermore, MajorMark abandons simple frequency counting in favor of a **clustering-based decoding strategy**. This method exhibits lower sensitivity to the Green List ratio; even when γ is large (implying the watermark signal is diluted), it can accurately separate the embedded message via cluster centers. Consequently, it maintains high-precision decoding capabilities despite variations in Green List size, eliminating the dependency on specific frequency thresholds.

2.3.3 BiMark (Unbiased Multilayer Watermarking)

As the primary baseline for this project, the **BiMark** scheme [6] proposed by Feng et al. represents the state-of-the-art in unbiased watermarking technology, effectively fusing the high generation quality of unbiased methods with the substantial payload

capacity of multi-bit schemes. BiMark introduces a **"Bit-Flip"** unbiased reweighting mechanism, which achieves model-agnostic detection via a token-count-based bisection approach. This innovation overcomes the limitations of previous unbiased methods that relied on accessing model logits for detection.

Crucially, BiMark employs a **Multilayer** reweighting architecture. By superimposing multiple independent reweighting layers, it amplifies the watermark signal while simultaneously enabling multi-bit information embedding through an XOR-enhanced position allocation technique. This architecture seeks to establish an optimal equilibrium among text quality, detection capability, and information capacity. Empirical results indicate that on short text sequences (200 tokens), BiMark achieves an extraction accuracy surpassing that of MPAC by over 30%, while maintaining lower perplexity.

2.4 New Frontiers: Diffusion Language Models (DLMs)

While autoregressive watermarking technologies have matured significantly, a paradigm shift is underway in generative AI. **Diffusion Language Models (DLMs)** [31, 8, 18], operating as a non-autoregressive generative architecture, are demonstrating potential that transcends traditional models.

Research by Gloaguen et al. has keenly identified the inefficacy of existing autoregressive watermarking schemes when applied to DLMs. All aforementioned methods (KGW, DiPmark, BiMark) rely heavily on the **Autoregressive Assumption**: that the generation of x_t is strictly conditioned on $x_{t-1}, x_{t-2}, \ldots$. Conversely, the generation process of DLMs is characterized by iterative denoising, where the token generation order is arbitrary or even parallel. This implies that during the sampling of a current token, the "historical context" required for hash calculation may be undetermined or exist in a noisy state. Consequently, the direct application of KGW-style schemes, which necessitate fixed contexts, results in vanishingly weak watermark signals.

To address this challenge, Gloaguen et al. proposed a watermarking strategy [7] based on **"Expectation Maximization."** Instead of relying on a deterministic history, this method imposes the watermark upon the distributional expectation of the context. Utilizing **"Expectation Boost"** (enhancing tokens that are "Green" in expectation) and **"Predictive Bias"** (enhancing tokens that render their neighbors "Green"), the scheme encourages the model to generate sequences that not only satisfy immediate watermark constraints but also construct a favorable watermark context for surrounding tokens. This conceptual shift transforms watermarking from local sequence locking into a **global optimization of an energy function**, representing a critical breakthrough in the migration of watermarking technology toward non-autoregressive architectures. However, this work remains confined to zero-bit detection; multi-bit attribution within the realm of DLMs remains an open research gap.

Chapter 3

Problem Statement

Despite the significant strides achieved by BiMark and MajorMark, the domain of multi-bit watermarking continues to confront a severe **"Pareto efficiency" bottleneck**. Furthermore, while BiMark stands as the current state-of-the-art unbiased multi-bit scheme, its practical robustness remains inadequate against complex attacks. Crucially, as generative architectures evolve towards **Diffusion Models**, existing theoretical frameworks for unbiased watermarking face the risk of **becoming theoretically inapplicable**.

3.1 The "Capacity-Robustness-Quality" Trilemma of Multi-bit Watermarking

Current multi-bit schemes are constrained by a rigid **trilemma**. To increase **Capacity** (bits per token), schemes typically necessitate aggressive vocabulary partitioning (e.g., MPAC's subdivision). This drastically reduces the pool of available tokens, thereby impairing **Text Quality**. Conversely, expanding the Green List to preserve quality (as in MajorMark) dilutes the statistical signal, rendering the watermark fragile against editing attacks (**Insufficient Robustness**). Although BiMark theoretically achieves unbiasedness and multi-bit capacity via its multilayer bit-flip mechanism, it exposes significant deficiencies in resilience during practical deployment:

1. **Lack of Structural Error Correction Capability:**
   BiMark relies on near-uniform XOR position allocation for information embedding. However, attacker edits introduce "noise" into the voting process. In BiMark, since layers are linked via the same token x_t, a single modification to x_t simultaneously flips the voting results of all d layers associated with that token. This implies that errors are highly Correlated, rather than Independent and Identically Distributed (I.I.D.) noise. Attacks are prone to causing extraction errors in specific bit positions. While BiMark employs a voting mechanism to mitigate this, it lacks the support of structural Error Correction Codes (ECC) found in schemes like RSBH or RBC. When facing Burst Errors or systematic edits, a simple majority voting mechanism fails to provide sufficient recovery capability. How to seamlessly integrate ECC into BiMark's unbiased multilayer framework without causing excessive signal dilution due to increased payload length remains an open question. The primary motivation of this project was to investigate whether ECC could remedy this vulnerability.
2. **Complexity of Inter-layer Optimization:**

The multilayer architecture assumes inter-layer independence to maintain unbiasedness. However, as the number of layers increases, the influence of signal strength (δ) in deeper layers diminishes, rendering their effect on logit bias less potent than that of shallower layers. Furthermore, increasing the layer count imposes additional computational overhead on the watermarking algorithm. Merely enhancing robustness by stacking complex mechanisms (such as high-order ECC or deep-layer injection) often devolves into a "Capacity-Quality" Zero-Sum Game.

3.2 The Research Gap: Watermarking for Diffusion Language Models

However, a more pressing concern lies in the structural incompatibility between existing methodologies and emerging architectures. As corroborated by Gloaguen et al., while **Diffusion Language Models (DLMs)** possess significant advantages in controllability and parallel generation, their security infrastructure remains virtually nonexistent.

All mainstream watermarking schemes (KGW, BiMark, MajorMark) [14, 6, 30] are predicated on the probabilistic chain rule $P(x_t|x_{<t})$. They postulate that during the generation of x_t, the history $x_{<t}$ is fixed and immutable, thereby serving as a reliable hash seed. In contrast, within DLMs, generation constitutes a global denoising process; tokens at positions t and $t+1$ may be determined simultaneously or non-sequentially. Consequently, the "history" remains fluid and indeterminate throughout the generation process. Direct application of ARLM hashing mechanisms precipitates a **"chicken-and-egg" causality dilemma**: the watermark seed is dependent upon a context that has yet to be finalized.

Although Gloaguen et al. introduced the inaugural DLM watermarking scheme [7], it is inherently **Biased**, introducing distributional distortion. Currently, there is a complete absence of research regarding **Unbiased** or **Distribution-preserving** watermarking for DLMs. Integrating the unbiased reweighting principles of BiMark or DiPmark into the global energy optimization framework of DLMs represents a highly challenging and unexplored frontier.

3.3 Research Objectives

Based on the preceding analysis, the objectives of this research project are organized into two evolutionary phases:

1. **Phase I (Completed/Current):** To conduct rigorous experimentation on the **BiMark** architecture, attempting to surmount its robustness bottlenecks via the integration of Error Correction Codes (ECC) and adaptive layering strategies. Subsequently, to analyze the underlying causes of the observed negative results, thereby establishing the inherent limitations of autoregressive unbiased watermarking.
2. **Phase II (Future/COMP9993):** To pivot the strategic focus towards **Diffusion Language Models (DLMs)** and develop a novel unbiased watermarking framework tailored for non-autoregressive generation processes.

This will entail adapting BiMark's "Bit-Flip" mechanism to the "Expectation Boost" framework of DLMs, effectively bridging the current technical void in unbiased watermarking for diffusion architectures.

Chapter 4

Methodology

This chapter details the experimental methodology adopted during the COMP9991 research phase. The core objective was to reproduce the **BiMark** framework and rigorously validate two hypotheses for improvement: (1) that the **Integration of Error Correction Codes (ECC)** enhances robustness; and (2) that **Layer-wise Dynamic δ Intensity Modulation** can reduce Perplexity (PPL) while preserving detection rates.

4.1 Baseline Establishment: Reproduction of BiMark

To establish a comparative baseline for subsequent experiments, we first rigorously reproduced the BiMark architecture. The core workflow comprises two sequential phases: embedding and extraction.

In the **Embedding Phase**, the system accepts a message $m \in \{0,1\}^l$ and a prompt x_p as input, employing a multilayer bit-flip reweighting function $R_{\theta,e}$. For each token generated, a context-seeded pseudo-random function determines a message index p and a one-time mask b. Subsequently, the target bit is encoded into a "fair coin flip" signal via the XOR operation $e^i = m[p] \oplus b^i$, which directs the reweighting of the vocabulary partition $(\mathcal{V}_0, \mathcal{V}_1)$. This process is superimposed across d independent layers, thereby accumulating statistical significance while preserving text quality.

Correspondingly, in the **Extraction Phase**, a voting matrix $M \in \mathbb{R}^{l\times 2}$ is constructed to accumulate votes for each bit position. For each detected token, the system estimates the flip state $\hat{e}^i$ based on the token's assignment within the layer partitions. Combining this with the reconstructed mask b^i, it performs a reverse XOR operation to cast a vote for the corresponding message bit $m[p]$. Finally, the original message is accurately reconstructed via a majority voting mechanism.

4.2 Experiment Design 1: Integration of Error Correction Codes (ECC)

To address the issue of insufficient robustness, and drawing inspiration from **RSBH** and **RBC**, an Error Correction Code (ECC) module is integrated prior to the BiMark embedding stage. The primary objective of this experiment is to investigate whether the introduction of algorithmic redundancy can enhance message extraction rates under rewriting attacks. We hypothesize that encoding the watermark message prior to embedding will enable the detector to correct bit-flip errors induced by text tampering.

- **Encoding Logic:** Instead of directly embedding the raw message m (e.g., 16 bits), the message is first encoded into a codeword c using Reed-Solomon (RS)

codes [23] (n, k) and Hamming codes (7, 4) [11]. This codeword c subsequently serves as the effective payload for the BiMark embedder.

- **Decoding Layer:** The system first utilizes BiMark's voting mechanism to extract a noisy bit string, denoted as $m'_{extracted}$, which is then passed to the decoder for error correction attempts.
- **Selection Rationale:** RS codes excel in handling Burst Errors, aligning closely with text watermarking scenarios where contiguous text segments are modified or deleted. Conversely, Hamming codes offer an effective mechanism for mitigating random, sparse single-bit errors (e.g., the substitution of a non-semantic word) while minimizing redundancy overhead.

Specifically, across varying message lengths of 8, 16, and 32 bits, we compared the baseline BiMark (without error correction), the linear block code Hamming(7,4), and the more robust Reed-Solomon (RS) code. For the RS codes, the error correction parameter was set to 2 (2 bytes), aiming to strike a balance between coding efficiency and error correction capability.

4.3 Experiment Design 2: Adaptive Strength (δ) Modulation

Standard BiMark applies a constant reweighting factor δ across all layers. We hypothesized that employing a **Variable** δ—specifically, assigning a distinct weighting factor δ_i to each layer $i \in [1, d]$—could further reduce PPL while maintaining detection rates.

We evaluated two non-uniform modulation schemes, **"Increasing"** and **"Decreasing,"** across varying depths ranging from 1 to 20 layers. The **Increasing Mode** (where δ rises linearly from 0.5 to 1.5) is designed to preserve the text generation quality of dominant partitions. Conversely, the **Decreasing Mode** (where δ descends linearly from 1.5 to 0.5) attempts to establish a strong robustness signal in shallow layers, subsequently mitigating noise interference via fine-tuning in deeper layers.

4.4 Datasets and Evaluation Framework

The experimental framework utilizes the **"RealNewsLike"** subset of the **C4 (Colossal Clean Crawled Corpus)** dataset [22] as prompts to simulate authentic news generation scenarios. The core generative model selected is **Llama-3.1-8B** [9]. For attack simulation, a pipeline was established using the **'dipper-paraphraser-xxl'** model [15] to evaluate watermark robustness against paraphrasing attacks via the semantic rewriting of generated text. Consistent with the original BiMark implementation, a window size of $h = 2$ and a fixed global seed of '42' were employed to ensure the reproducibility of both partition generation and message encoding.

4.5 Attack Simulation and Evaluation Metrics

To comprehensively evaluate the robustness of the watermark, we constructed an automated multi-level rewriting attack pipeline utilizing the **DIPPER** model [15] ('kalpeshk2011/dipper-paraphraser-xxl'). The attack intensity is jointly defined by **Lexical Diversity (L)** and **Order Diversity (O)**, covering a full spectrum ranging from no attack (Level 0) to high-intensity rewriting (Level 60, where $L = 60, O = 60$).

Regarding evaluation metrics, this study focuses primarily on **Message Extraction Rate (MER)** and **Perplexity (PPL)**. MER, defined as $1 - \text{BER}$ (Bit Error Rate), quantifies the capability to successfully recover watermark information under specific attack intensities and varying text lengths (50 to 300 Tokens). Simultaneously, to ensure that watermark embedding does not significantly impair the fluency of the generated text, we employed the larger-parameter **Gemma-2-9B** model [25] as an **Oracle** to calculate the PPL values. By analyzing the variation curves of MER and PPL across different attack intensities, we can qualitatively assess the trade-off between robustness and stealthiness achieved by each proposed improvement scheme.

Chapter 5

Preliminary Results and Analysis

This section provides an in-depth analysis of the enhancement experiments conducted on BiMark. Contrary to initial hypotheses, the integration of Error Correcting Codes (ECC) and the adoption of a variable layer intensity strategy did not improve performance under specific constraints; instead, they led to degradation in certain metrics.

5.1 Failure Analysis of Simple ECC Integration

Although Reed-Solomon codes have proven to be a critical pillar of robustness in RSBH and RBC, the direct integration of ECC within BiMark's architecture resulted in a significant decline in Bit Accuracy (BA) for short text scenarios ($T <$ 200 tokens). For instance, using RS(4, 2) to encode a 16-bit payload yielded a lower recovery rate than directly embedding 16 bits of raw data.

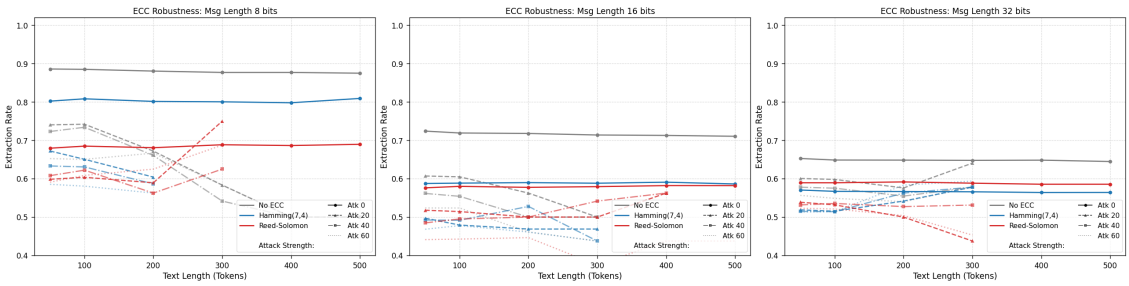

Figure 5.1: The experiments utilized three different watermark message bit lengths, as illustrated in the figure. Different colors represent different ECC algorithms, while varying color intensities and line styles indicate different attack intensities. The algorithm specifies a maximum allowable length; consequently, under high-intensity attacks, some generated texts are prematurely truncated.

Theoretical Root Cause Analysis:

- **Signal Dilution via Redundancy:** BiMark pseudo-randomly assigns tokens to specific bit positions in the message for embedding via a hashing mechanism. This is a random sampling process. When ECC is introduced, the message length expands from k (payload) to n (codeword)—for example, from 16 to 32. Under the constraint of a fixed text length T, the average number of tokens allocated to each bit position (i.e., the "number of votes") is diluted by a factor of k/n. BiMark's extraction logic relies on the Law of Large Numbers: the more votes per bit position, the higher the statistical significance, and the lower the error rate. When the number of votes falls

below a critical threshold, the Raw Bit Error Rate (RBER) for a single bit rises sharply.

- **Avalanche Effect:** ECC (such as RS codes) has an explicit error correction bound of $(n-k)/2$. If the RBER exceeds this bound, decoding will not only fail but might also produce a result worse than random guessing due to incorrect parity calculation. In the short text scenario, the RBER spike caused by signal dilution directly breaches the ECC's correction threshold, leading to complete decoding failure. This demonstrates that: **in finite-length texts, simply increasing coding redundancy cannot compensate for the degradation in the Signal-to-Noise Ratio (SNR) caused by sample sparsity.**

5.2 Analysis of Layer δ Variance Failure

Attempts to optimize the signal by varying the inter-layer δ value also failed to improve the PPL-BA trade-off curve.

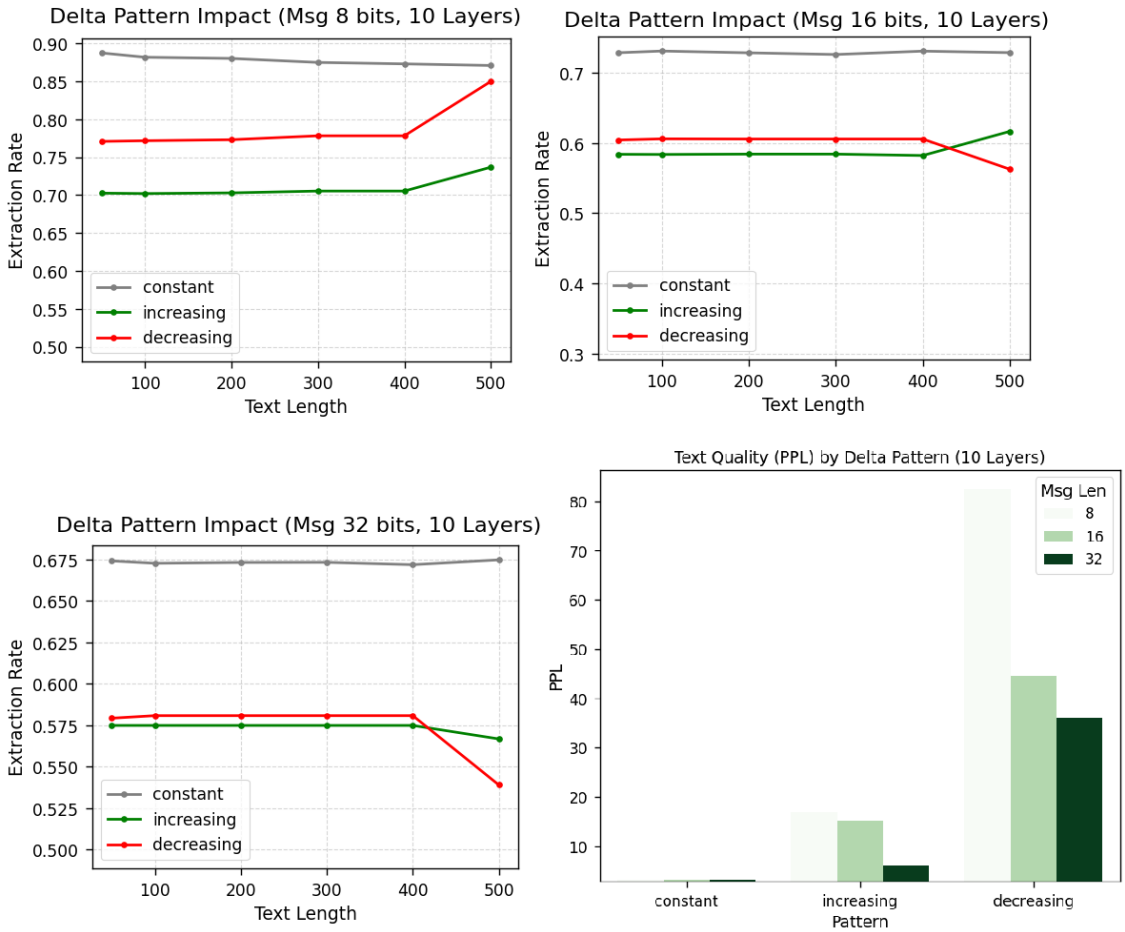

Figure 5.2

By observing the fixed intensity ($\delta = 1$) baseline and high-variance configurations (where δ is incrementally increased/decreased between 0.5 and 1.5) for 8-bit,

16-bit, and 32-bit payloads, it was found that configurations with high variance exhibited not only lower bit extraction rates but also higher Perplexity (PPL) compared to the uniform δ baseline.

Theoretical Root Cause Analysis:

Ablation studies on BiMark indicate that shallow layers (i.e., the first few layers) have a dominant influence on the final probability distribution. This is because the token probability distribution is relatively uniform in the initial stages, making the re-weighting operations most effective then. Simultaneously, if δ is too large, it can cause the probabilities of certain tokens to rapidly "collapse" toward zero in the early layers. Once a probability becomes zero in an early layer, no subsequent layer (regardless of its δ) can recover it.

1. Therefore, when adopting an **increasing** strategy, the most critical first few layers use the weakest intensity ($\delta = 0.5$). This means the algorithm minimally intervenes in the probability distribution during the "golden window" of watermark embedding. By the time δ increases to 1.5 in later layers, this strong intervention can barely improve the extraction rate, yet it continues to compromise text fluency by forcibly altering the probability distribution. The continuously increasing δ intensity at this point is almost equivalent to introducing noise, leading to higher PPL.
2. Conversely, when adopting a **decreasing** strategy, $\delta = 1.5$ might cause a **Probability Collapse**, forcing the probabilities of a large number of candidate tokens to zero in layers 1-2. This excessive intervention may destroy the ability of subsequent layers to fine-tune the distribution, effectively reducing the number of layers that can carry the watermark. This essentially "kills" the ability of these tokens to carry subsequent layer watermark information, leading to a decrease in the extraction rate. Moreover, if a layer uses an extremely high δ to flip token probabilities, it may push the sampling towards the tail of the distribution. Even if the "expectation" is maintained, the "variance" of the sampling process is significantly increased. High PPL is often a severe penalty for selecting low-probability tokens, thus the increase in variance directly leads to a degradation in the text quality metric.

5.3 Preliminary Experimental Conclusion

The failure of the aforementioned enhancement strategies highlights the delicate balance mechanism within the BiMark framework. BiMark's efficiency is built upon the simplicity of its structure and statistical laws. Under constrained text lengths, introducing structural complexity (ECC) or parameter variance (δ scheduling) disrupted this equilibrium. This also suggests that continuing to optimize BiMark within the Auto-Regressive (ARLM) framework faces the dilemma of diminishing marginal returns. To achieve a qualitative leap, the research must pivot to a brand-new generation paradigm—**Diffusion Models**—where the optimization space is not yet filled by such limitations.

Chapter 6

Future Plan (COMP9993): Watermarking Research in Diffusion Language Models

Given the optimization bottleneck for unbiased watermarking in auto-regressive models, the focus of this research project in the COMP9993 phase will shift to the emerging field of Diffusion Language Models (DLMs). The goal is to pioneer a robust and unbiased watermarking scheme specifically tailored for the non-auto-regressive generation process of DLMs.

6.1 Theoretical Basis and Challenges

As established by Gloaguen et al. [7], there is a fundamental incompatibility between ARLM watermarking and DLMs, primarily centered on the **absence of a fixed history**. In the **ARLM paradigm**, $x_{<t}$ is immutable when sampling x_t. Conversely, in the **DLM paradigm**, the sequence $x_{1:T}$ is an overall iterative evolution from noise to text.

If a hash function $h(x_{context})$ is simply applied to determine the green list for x_t, the watermarking logic will fail because the context tokens may change (be re-masked or re-sampled) in subsequent diffusion steps. Experiments show that directly applying ARLM watermarking to DLMs results in a near-zero detection rate, as the watermark's "anchor" (the context) is unstable.

6.2 Proposed Methodology: Unbiased Expectation Maximization for DLMs

This research proposes adapting the "unbiased reweighting" principle from BiMark/DiPmark into the "Expectation Boost" framework proposed by Gloaguen et al [7].

6.2.1 Core Concept: Global Energy Optimization

We view watermarking as a global energy optimization problem, defining the energy function $J(x) = \sum_t \mathbb{I}(x_t \in GreenList(x_{context}))$. The goal is to maximize the sequence's overall "greenness." The objective is to algorithmically sample a distribution $Q(x)$ that maximizes $\mathbb{E}_{x\sim Q}[J(x)]$ while minimizing the divergence $KL(Q||P)$ from the original DLM distribution P.

6.2.2 Innovation: Unbiased Diffusion Reweighting

Gloaguen et al [7]. currently use a biased Logit Tilt: $q_t \propto p_t \exp(\delta\alpha_t)$, which introduces distortion. We will investigate applying **Bit-Flip Reweighting** or **Correlated Binary Sampling Channels (CBSC)** during the diffusion steps.

The specific implementation idea is as follows: In diffusion step k, the model predicts the probability distribution $p_t^{(k)}$ for the masked token. Instead of shifting the Logits (which introduces bias), we partition the vocabulary V into V_0, V_1 based on the hash value of the current estimate of the context. We calculate the expected hash value of the context token distribution within a window of size k. If the context is masked, we calculate the hash for the top m candidates for that context position to get a prediction for that position. Finally, by applying reweighting at each diffusion step that satisfies $\mathbb{E}[Q] = P$, we hypothesize that the resulting generated sequence will maintain high quality (unbiased) while converging to a state with high watermark energy.

6.3 Implementation Plan

1. **Model Selection**: Utilize open-source DLMs, such as **LLADA-8B** [18] or **DREAM-7B** [31], which serve as standard benchmarks in the field of diffusion generation.
2. **Baseline Construction**: Replicate Gloaguen et al.'s "Expectation Boost" watermark to establish a biased baseline.
3. **Algorithm Development**:
   - Adapt BiMark's reweighting function to operate on the "Meta-Logits" of the diffusion process.
   - Develop a "soft" hashing mechanism capable of accepting **probabilistic context (distributions)** rather than discrete tokens, allowing the watermark to influence the diffusion trajectory before tokens are definitively finalized.
4. **Evaluation**:
   - Compare against biased DLM watermarks and post-hoc ARLM baselines.
   - Metrics: Detection Rate (TPR@FPR), Quality (GPT-4o Judge, PPL), and Robustness against Context-aware substitutions.

6.4 Expected Outcomes

This research aims to produce the first **unbiased** watermarking scheme suitable for Diffusion Language Models. If successful, this will represent a significant contribution to the field, offering a solution that combines the high-quality generation capabilities of DLMs with rigorous provenance guarantees, while being robust against the non-sequential nature of modern generative AI.

References

[1] Jack T Brassil, Steven Low, Nicholas F Maxemchuk, and Lawrence O'Gorman. Electronic marking and identification techniques to discourage document copying. *IEEE Journal on Selected Areas in Communications*, 13(8):1495–1504, 1995.

[2] Patrick Chao, Yan Sun, Edgar Dobriban, and Hamed Hassani. Watermarking language models with error correcting codes, 2025.

[3] Ruibo Chen, Yihan Wu, Yanshuo Chen, Chenxi Liu, Junfeng Guo, and Heng Huang. A watermark for order-agnostic language models, 2024.

[4] Ruibo Chen, Yihan Wu, Junfeng Guo, and Heng Huang. Improved unbiased watermark for large language models, 2025.

[5] European Commission. Regulation (eu) 2024/1689 of the european parliament and of the council of 13 june 2024. Technical report, European Commission, 2023.

[6] Xiaoyan Feng, He Zhang, Yanjun Zhang, Leo Yu Zhang, and Shirui Pan. Bimark: Unbiased multilayer watermarking for large language models, 2025.

[7] Thibaud Gloaguen, Robin Staab, Nikola Jovanović, and Martin Vechev. Watermarking diffusion language models, 2025.

[8] Shansan Gong, Shivam Agarwal, Yizhe Zhang, Jiacheng Ye, Lin Zheng, Mukai Li, Chenxin An, Peilin Zhao, Wei Bi, Jiawei Han, Hao Peng, and Lingpeng Kong. Scaling diffusion language models via adaptation from autoregressive models. *International Conference on Learning Representations*, 2025.

[9] Aaron Grattafiori et al. The llama 3 herd of models, 2024.

[10] Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, et al. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.

[11] Richard W Hamming. Error detecting and error correcting codes. *Bell System technical journal*, 29(2):147–160, 1950.

[12] The White House. Executive order on the safe, secure, and trustworthy development and use of artificial intelligence. Technical report, The White House, USA, 2023.

[13] Zhengmian Hu, Lichang Chen, Xidong Wu, Yihan Wu, Hongyang Zhang, and Heng Huang. Unbiased watermark for large language models, 2023.

[14] John Kirchenbauer, Jonas Geiping, Yuxin Wen, Jonathan Katz, Ian Miers, and Tom Goldstein. A watermark for large language models. In *International Conference on Machine Learning*, pages 17061–17084. PMLR, 2023.

[15] Kalpesh Krishna, Yixiao Song, Marzena Karpinska, John Wieting, and Mohit Iyyer. Paraphrasing evades detectors of ai-generated text, but retrieval is an effective defense, 2023.

[16] Minjia Mao, Dongjun Wei, Zeyu Chen, Xiao Fang, and Michael Chau. Watermarking low-entropy generation for large language models: An unbiased and low-risk method, 2025.

[17] Hasan Mesut Meral, Bülent Sankur, A Sumru Özsoy, Tunga Güngör, and Emre Sevinç. Natural language watermarking via morphosyntactic alterations. *Computer Speech & Language*, 23(1):107–125, 2009.

[18] Shen Nie, Fengqi Zhu, Zebin You, Xiaolu Zhang, Jingyang Ou, Jun Hu, Jun Zhou, Yankai Lin, Ji-Rong Wen, and Chongxuan Li. Large language diffusion models. *arXiv preprint arXiv:2502.09992*, 2025.

[19] Lip Yee Por, Kok Sheik Wong, and Kuang Ooi Chee. Unispach: A text-based data hiding method using unicode space characters. *Journal of Systems and Software*, 85(5):1075–1082, 2012.

[20] Wenjie Qu, Wengrui Zheng, Tianyang Tao, Dong Yin, Yanze Jiang, Zhihua Tian, Wei Zou, Jinyuan Jia, and Jiaheng Zhang. Provably robust multi-bit watermarking for ai-generated text, 2025.

[21] Alec Radford, Karthik Narasimhan, Tim Salimans, and Ilya Sutskever. Improving language understanding by generative pre-training. Preprint, 2018.

[22] Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of machine learning research*, 21(140):1–67, 2020.

[23] Irving S Reed and Gustave Solomon. Polynomial codes over certain finite fields. Technical Report 202, Lincoln Lab., Massachusetts Inst. Technol., Lexington, MA, 1960.

[24] Stefano Giovanni Rizzo, Flavio Bertini, and Danilo Montesi. Content-preserving text watermarking through unicode homoglyph substitution. In *Proceedings of the 20th International Database Engineering & Applications Symposium*, pages 97–104, 2016.

[25] Gemma Team et al. Gemma 2: Improving open language models at a practical size, 2024.

[26] Umut Topkara, Mercan Topkara, and Mikhail J Atallah. The hiding virtues of ambiguity: quantifiably resilient watermarking of natural language text through synonym substitutions. In *Proceedings of the 8th Workshop on Multimedia and Security*, pages 164–174, 2006.

[27] Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, et al. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023.

[28] Lean Wang, Wenkai Yang, Deli Chen, Hao Zhou, Yankai Lin, Fandong Meng, Jie Zhou, and Xu Sun. Towards codable watermarking for injecting multi-bits information to llms, 2024.

[29] Yihan Wu, Zhengmian Hu, Junfeng Guo, Hongyang Zhang, and Heng Huang. A resilient and accessible distribution-preserving watermark for large language models, 2024.

[30] Jiahao Xu, Rui Hu, and Zikai Zhang. Majority bit-aware watermarking for large language models, 2025.

[31] Jiacheng Ye, Zhihui Xie, Lin Zheng, Jiahui Gao, Zirui Wu, Xin Jiang, Zhenguo Li, and Lingpeng Kong. Dream 7b: Diffusion large language models, 2025.
[32] KiYoon Yoo, Wonhyuk Ahn, and Nojun Kwak. Advancing beyond identification: Multi-bit watermark for large language models, 2024.
[33] Xuandong Zhao, Prabhanjan Ananth, Lei Li, and Yu-Xiang Wang. Provable robust watermarking for ai-generated text, 2023.